CS383 · SOFTWARE ENGINEERING

Lecture 7 Software Requirements

Midterm summary

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01 What is a Requirement?

A **Software requirement** answers one question: what is the software expected to do?

Key idea

The whole lecture is two classifications. Every requirement sentence gets one name from each classification.

Classification	Based on what?	Options
First	Level of detail	**User** or **System**
Second	Type of content	**Functional** or **Non-functional**

02 User vs System requirements

	User requirement	System requirement
Language	Plain language, in the user's own words.	A detailed and precise description.
Level	General and brief (High-level).	States exactly what will be built.
Reader	Does not care how the system is implemented.	Wants to know exactly what the system does.

The **SRS** document must include both types. We write it at two levels because its readers have different backgrounds.

How do you tell them apart in the exam?

USER **A general sentence with no numbers or details**

"The system should be easy to use for first-year students."

SYSTEM **A specific sentence with precise details**

"The interface will display student GPA to two decimal places."

03 Functional vs Non-functional

Type	Meaning
Functional (FR)	A service or function the system provides. In short: **what it does**. It may also state what the system should **not** do.
Non-functional (NFR)	A constraint or quality on the system as a whole. In short: **how** it works and how well.

Clue words

If you see this in the sentence	Type
allow · create · display · retrieve · export · update	**FR**
fast · easy to use · secure · reliable · encrypted	**NFR**
within 1 second · 100 concurrent users · 99.9%	**NFR**
A law, a standard, a programming language, or a required platform	**NFR**

Watch out

The word **shall** does not make a sentence Functional. This sentence is an **NFR** because fast is a quality attribute:

> "The report generation shall be fast."

- An **NFR** can be more important than an **FR**. Without it, the system may not work at all.
- Requirements must be **Complete** (nothing is missing) and **Consistent** (no contradictions).

04 The three types of Non-functional requirements

To find the type, ask: **who imposed this constraint?**

Type	Who imposed it?	Subtypes	Example
Product	The nature of the product itself.	Usability Efficiency (Performance, Space) Dependability Security	respond in ≤ 300 ms
Organization	The customer's company or the developer's company.	Environmental Operational Development	must be built in Java 21
External	An outside party: a law, a standard, ethics.	Regulatory Ethical Legislative (Accounting, Safety/Security)	must comply with HIPAA

05 How do we measure an NFR?

A word like fast cannot be tested. To make it testable, we tie it to a measure (**Metric**).

Property	How it is measured
Speed	Number of transactions processed, response time.
Size	Megabytes, number of storage devices.
Ease of use	Training time.
Reliability	Mean time to failure, number of failure occurrences.
Robustness	Time to restart after a failure, percentage of data lost on failure.
Portability	Number of systems it runs on.

06 Solved example from the slides

A hospital asks for a system to manage its patient records. The system shall allow the creation of new records for newly registered patients using national IDs. It shall allow retrieving patient information using IDs, names, or phone numbers. The system should be operational under the JVM using JDK 17 and above. It should respond to operations in less than 1 second. Patient information should be end-to-end encrypted.

Requirement	Type
Create a new record for a newly registered patient using the national ID.	**FR**
Retrieve patient information by ID, name or phone number.	**FR**
Respond in less than 1 second.	**NFR · Product**
Patient information is end-to-end encrypted.	**NFR · Product**
Run on the JVM with JDK 17 and above.	**NFR · Organization**

07 The SRS document

The **Requirement specification** is the detailed documentation of the user and system

requirements.

Ways to write a Requirement

Method	What it is
Natural language	Ordinary sentences. Each sentence holds one requirement.
Structured natural language	Plain language in an agreed template. **The most widely used.**
Design description language	Similar to a programming language. Rarely used.
Graphical notations	Diagrams and models such as UML.
Mathematical specification	Mathematical notation such as sets and finite-state automata.

Every Requirement must include

`Description · Rationale · Inputs · Outputs`

`Actions · Pre-conditions · Post-conditions`

Word	Meaning
shall	Mandatory.
should	Desirable.

08 Requirements management

Requirements will certainly change, because the technical and business environment changes, and because the paying customer imposes new requirements.

The **Customer** and the **End user** are not always the same person: one pays, the other uses the system.

What must a management tool provide?

#	Function	What it does
1	**Requirement storage**	Keeps all the requirements in one central, secure place.
2	**Change management**	Analyzes the problem, then works out the cost of the change, then carries it out.
3	**Traceability management**	Finds the other requirements affected by the change.

09 How does it appear in the midterm?

This lecture appeared in the midterms, and in one of them it was worth 10 marks out of 20.

Question 1: User/System and FR/NFR (4 marks)

Sentence	User / System	FR / NFR
"The system should be easy to use for first-year students."	User	NFR
"The application shall allow password reset using a verified email."	System	FR
"The report generation shall be fast."	User *	NFR
"The interface will display student GPA to two decimal places."	System	FR

* This cell is blank in the answer key. User fits because the statement is general and has no number.

Question 2: Product / Organization / External (6 marks)

Sentence	Answer
The API shall respond in ≤ 300 ms (p95) for read requests under 100 concurrent users.	**Product**
All new services must be built in Java 21 and deployed on the organization's Kubernetes platform.	**Organization**
Patient data must comply with HIPAA and local privacy law X.	**External**
The mobile app shall support offline read-only access for 24 hours.	**Product**
The system shall export reports in CSV and PDF.	**Product**
Device firmware must interoperate with ISO/IEC 14443 Type A NFC readers.	**External**

Question 3: Extract 2 FR and 2 NFR (4 marks)

A company needs a Task Management System. The system will allow employees to create tasks with a title, description, priority level and due date, and assign these tasks to members. Employees can update the status of their tasks as they work on them. The system should handle up to 100 concurrent users without delays and ensure that user data is encrypted during transmission.

Type	Answer
FR	The system will allow employees to create tasks and assign them to members.
FR	Employees can update the status of their tasks.
NFR	The system should handle up to 100 concurrent users without delays.
NFR	User data is encrypted during transmission.

10 Common mistakes

Mistake	Correct
Treating any sentence with shall as an FR	Look at the content. A quality or a numeric limit is an NFR.
Mixing up the two classifications	User/System is about detail. FR/NFR is about content.
Putting a law under Organization	Laws and outside standards are **External**.
Putting "must use Java" under Product	A required language or platform is **Organization**.
Swapping shall and should	**shall** is mandatory, **should** is desirable.

11 Test yourself

1. Classify (User/System and FR/NFR): **The system should be reliable for daily use.**
2. Classify (User/System and FR/NFR): **The system shall send an SMS code when the user logs in.**
3. Which type of NFR: **All source code must follow the company's coding standard.**
4. Which type of NFR: **User data handling must comply with the national data protection law.**
5. What is the most widely used way to write requirements?

Answers

#	Answer	Reason
1	User · NFR	A general sentence, and reliable is a quality attribute.
2	System · FR	A specific service with its details.
3	**Organization**	A development rule set by the company itself.
4	**External**	Imposed by a law.
5	**Structured natural language**	Stated in the slides.